

PhysBench: A Verifiable Multi-Domain Benchmark for Physics-Informed Machine Learning

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Abstract—Benchmarks for physics ML mostly share a structural weakness: the ground truth and the model are validated against the same simulation code, so agreement can reflect shared error rather than physical correctness. PhysBench is a benchmark of twelve physics domains built on the independent-verifier principle: every target is produced by a closed-form or high-resolution numerical solution implemented separately from the training signal, and every prediction is scored against an independent governing-equation residual, not the generating code. We provide a committed 75-run baseline matrix (5 domains $\times$ 3 architectures $\times$ 5 seeds) and an operator-network comparison on the ten trajectory domains. Held-out error spans two orders of magnitude across domains, and the best architecture differs by domain. PhysBench is small by design (a few hundred samples per domain, CPU-minutes): it is built to expose mechanisms, not to win leaderboards.

Index Terms—benchmark; physics-informed machine learning; verification; neural operators

I. INTRODUCTION

A benchmark claim – ‘this model gets the physics right’ – is only meaningful if ‘right’ is defined by something the benchmark author did not also write. PhysBench enforces two independence conditions: targets are generated separately from the training signal, and predictions are scored against the governing equation by verifiers written from a different formulation (finite differences, Euler/RK4, energy balance).

II. DOMAINS

Twelve domains: algebraic closed-form trajectories, energy conservation, harmonic and relaxation ODEs (spring, LC, damped oscillator), central-force motion (Kepler), fourth-order bending (beam, cantilever), linear drag, a conservation-law PDE (Burgers), and an elliptic field problem (heat plate). Each provides a parameterized instance family, exact answers, shape-normalized targets, and a deterministic, seeded data generator.

III. RESULTS

The 75-run baseline matrix is fully committed (physx/models/matrix/). Held-out error spans two orders of magnitude; the best architecture differs by domain. On the ten trajectory domains, ten dedicated DeepONets reach median held-out error 0.037 vs. 0.110 for a single law-conditioned generalist, which beats the specialists outright on spring (0.111 vs. 0.122) and LC (0.100 vs. 0.192). The benchmark’s point is not that one architecture wins – it is that the right question is per-domain and per-capacity, and independent verifiers let both statements be made precisely.

REFERENCES

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